

Optical tweezers for trapping in a microfluidic environment

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Optical tweezers use the force from a light beam to implement a precise gripping tool. Based purely on an optical principle, it works without any bodily contact with the object. In this paper we describe an optical tweezers that targets an application within the framework of nuclear magnetic resonance (NMR) spectroscopy of small objects, which are embedded inside a microfluidic channel that will be integrated in a micro-NMR detector. In the project's final stages, the whole system will be installed within the wide bore of a superconducting magnet. The aim is to precisely maintain the position of the object to be measured, without the use of susceptibility disturbing materials or geometries. In this contribution we focus on the design and construction of the tweezers. For the optical force simulation of the system we used a geometrical optics approach, which we combined with a ray fan description of the output beam of an optical system. By embedding both techniques within an iterative design process, we were able to design efficient optical tweezers that met the numerous constraints. Based on details of the constraints and requirements given by the application, different system concepts were derived and studied. Next, a highly adapted and efficient optical trapping system was designed and manufactured. After the components were characterized using vertical scanning interferometry, the system was assembled to achieve a monolithic optical component. The proper function of the optical tweezers was successfully tested by optical trapping of fused silica particles. © 2018 Optical Society of America

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1. INTRODUCTION

Optical tweezers (OT) enable the manipulation of microscopic objects without bodily contact. Thus, objects can first be trapped, then, e.g., moved or rotated along complex trajectories, after which they can be released at will. The absence of adhesive forces between the object and a gripper makes the OT a powerful tool at the microscale and nanoscale. Since their invention almost half a century ago by Ashkin *et al.* [1,2], an immense effort has been and still is being invested to employ optical trapping for an increasing number of applications [3–6]. For example, subsequent reports focus on increasing the trapping efficiency (ratio of laser power to optical trapping force and trap stiffness) [7–10], the positioning accuracy of the optical trap [6,11], the introduction of torque [12,13], or the enhancement of observation and detection methods [14–16].

The principle of nuclear magnetic resonance (NMR) spectroscopy rests on the manipulation of spin degrees-of-freedom

of nuclei (most frequently investigated: protons ^1H and carbon-13 ^{13}C). A sample is placed inside a homogeneous high and static magnetic field B_0 , which causes the spin energy levels of the nuclei to split—the so-called Zeeman effect—and the populations to redistribute among these levels. By applying an additional pulsed RF magnetic field $B_1 \perp B_0$, the net magnetization is deflected and precesses in B_0 at the Larmor frequency $f = \gamma B_0$ (γ is the gyromagnetic constant of the nucleus), thereby relaxing back to an aligned orientation along B_0 . The radiation emitted by the precessing magnetization is detected and frequency encodes information about the nuclear levels as a function of the molecular neighborhood. For more information please see [17–20].

Our ultimate goal is to apply NMR microscopy to small biological objects, such as cells, cell clusters, or oocytes. These usually have a refractive index ranging from 1.35 to 1.4 [21,22] and are transparent to radiation in the NIR (e.g., 1070 nm) wavelength range. The form and orientation

of the targeted cells can be approximated by spheroids (spherical, oblate, prolate, and needle-like shapes), which are commonly used objects in optical trapping [23–25]. In this paper we focus on the design and construction of a dedicated optical trap. It will ultimately be integrated and operated inside the bore of a high field superconducting magnet. This measurement technique requires a very homogeneous magnetic field distribution. Thus, great effort is spent, through careful NMR probe head design and the use of superconducting shim-coils, to achieve field homogeneities of 1 ppb or better, which the optical design should not disturb [26–28]. In addition, the position of the measurement object relative to the magnetic isocenter has to be preserved during the entire measurement time. The voxel diameter used for spectroscopic investigations will be of the order of 10–50 μm , so that displacements or oscillations of the sample will introduce noise and lower the spatial spectroscopic resolution. These seemingly disparate goals are hard to achieve with conventional setups but can potentially be facilitated by the use of a suitably configured optical trap.

For this purpose, the OT design had to fulfill several boundary conditions and requirements. Starting with the environment, the optical system had to be compatible with high magnetic fields, meaning that its components and mounting needed to be made of non-magnetic materials with similar susceptibility, such as copper or polymethylmethacrylate (PMMA). Next, the setup had to fit within the bore of a magnet (typically a diameter of 54–89 mm), and had to match the geometrical specification of the spectrometer equipment, such as the probe head, as well as the microfluidic platform containing the sample to be measured. This led to a customized optical system for the OT including three required optical functions: beam shaping, 90° beam deflection, and high numerical aperture (NA) focusing.

For the design of the optical system, an iterative optical design process was used [9,10]. It combined standard optical design software, a force simulation tool based on a geometrical optics approach (GOA), and a ray-fan-wise approximation of the output beam of the optical system. Based on the design process, an OT system was designed and toleranced. Once the components were manufactured, the optical system was assembled and characterized. Finally, we optically trapped spherical fused silica particles of 10 μm in diameter to verify the proper function of the OT.

The remainder of the paper is structured as follows: Section 2 briefly introduces the design procedure and the optical force simulation. The characteristics of the environment are clarified here as well. In Section 3, the optical concept, as well as the design of the OT, are discussed. Next, in Section 4, the manufactured optical components are characterized and the optical setup for optical trapping is explained in detail. The optical trapping is presented in Section 5. Experimental results are discussed and compared with the theoretical predictions achieved with the GOA-based force simulation. The paper ends with a conclusion and outlook in Section 6.

2. DESIGN APPROACH AND ENVIRONMENT

The basis for the design of the optical trap is a realistic prediction of the trapping force. This in turn depends on a

precise prediction of the light path to the point of trapping. In the case of the compact optical trap targeted here, many closely spaced optical elements and interfaces have to be traversed before the focused beam actually interacts with the particle. Here we first discuss the force simulation, and then we describe the optical environment that constrains the problem.

A. Optical Force Simulation

The method used for the optical force simulation is based on a GOA, valid for particles larger than the wavelength [29], and first applied to calculate optical forces by Ashkin [30].

The starting point of the GOA is a ray-based approximation of a light beam which interacts with the object. A beam of course consists of many rays, and each ray is described by a vector with an origin, a direction, and a magnitude. Additionally, each ray vector is linked to a second vector that tracks its polarization, as well as polarization direction changes after each interaction with an optical interface. If the ray interacts with the interface of an object, its intensity and direction of propagation will change. This can be physically associated with a force exerted by the ray on the object. By considering all rays, and summing up over all ray–surface interactions, the overall optical force acting on an otherwise freely suspended object can be calculated [9,31]. This force calculation routine is described in detail in an earlier published work [9].

To interface the optical force simulation with a commercially available optical design software tool (Zemax, OpticStudio [32]), the optical system is described in a simplified fashion. All rays exiting the optical system are stored as an array of pixels with a certain pixel size and pitch. The array defines the origin of each ray, while the distance to the focal point is used to determine the direction of propagation. The value of each pixel represents the intensity of the ray. The array of pixels therefore establishes the interface to the standard optical design software [32], which is necessary for the optical design process described next.

B. Iterative Optical Design Process

For the development of such a highly specialized optical trapping system, the intensity of the optical force has to act as a merit function in the optical design process. Thus, we established an iterative design process, which links the optical force simulation with the optical design software [32].

In our particular case, the optical system has to ensure the trapping of particles which are dispersed in a fluid. The fluid is carried by a microfluidic sample container that is positioned inside the bore of a superconducting magnet. The application defines the optical environment and function of the system to be designed, the characteristics of the particle to be trapped, and additional restrictions to be determined.

Based on the specifications and following the design process structure depicted in Fig. 1, characteristics of the OT will be derived and investigated. Next, the optical design software is used to design the initial system. If the system successfully generates an optical trap, its function is subsequently confirmed by the optical force simulation. In the next phase, the optical system is optimized in terms of improving the trapping performance (ratio of laser power to optical force), by modifying

maintained as completely homogeneous and parallel to the magnet axis for as long a distance as possible. The bore of the magnet has a diameter of 70 mm and a total length of 1500 mm. Due to the other non-optical components, such as the probe head, gradient coils, loading devices, cables and supply lines, etc., the optimum space available for the optical trapping setup has a semi-cylindrical shape with 70 mm in diameter and 100 mm in height.

Another major requirement is the compatibility of the optical system with the high magnetic field existing in the bore of the superconducting magnet. For the optical elements themselves this condition is not critical. Also, the optic holders are usually made of aluminum, which is compatible with high magnetic fields, but commercial lens mountings often are made of materials containing nickel, iron, etc., and hence should be avoided [26,28]. The complex layer structure of the sample container and the chip, the space restrictions for the optics, and the demands regarding suitable materials lead to a customized solution for the OT.

3. DESIGN PROCESS OF THE OT

The main requirements and constraints to be met during the design process are summarized in Table 1.

Based on these specifications, different concepts were derived. All concepts contain three optical functions: beam shaping, beam deflection, and focusing. In the last part of this section the design of the OT is presented and discussed.

A. Development of the Concept

As reported previously [10], the beam shaping of a Gaussian laser beam into an annular beam profile leads to higher optical trapping forces. This promising approach is pursued again for the current optical tweezers. In an annular beam, the high intensity points are arranged in a ring shape away from the optical axis. Thus, focusing the ring on a particle reduces incident rays with low angles and consequently reduces the scattering force. This leads to an increase of the net force of optical trapping.

Due to the limited space available for the OT, a 90° beam deflection is intended to reduce the system size in one dimension and to simplify the beam access along the bore of the magnet.

Table 1. Constraints and Requirements

Specification	Note
Function	Trapping of single object, no manipulation and observation planned.
Laser Source	$M^2 < 1.2$, wavelength 1070 nm, continuous wave.
Object to be trapped (cell, oocyte, bead)	Shape: spherical and spheroidal; Size: $\approx 10 \mu\text{m}$; Material: cell ($n = 1.35$ to 1.4), PMMA ($n_{1070 \text{ nm}} = 1.482$), fused silica ($n_{1070 \text{ nm}} = 1.450$).
Surrounding medium	Fluid (refractive index ≈ 1.33).
Outer environment (system size)	Bore of superconducting magnet, semi-cylindrical volume 70 mm $\times$ 30 mm $\times$ 100 mm (diameter, height, length).
Inner environment (working distance)	Microfluidic chip, 875 μm wall thickness plus 50 μm half channel width.

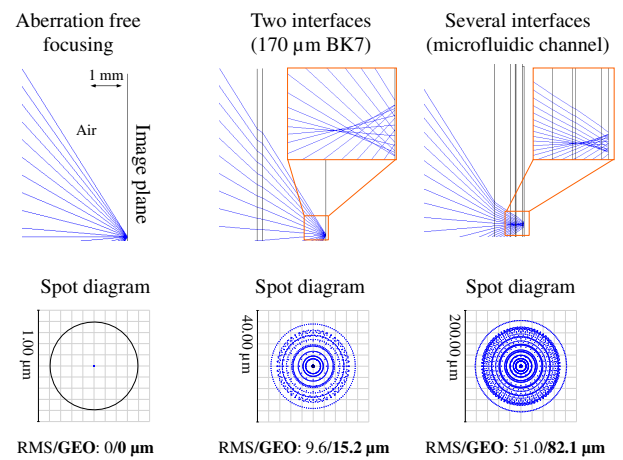


Fig. 4. Increase of spherical aberrations by introducing optical interfaces.

In order to verify the extent to which the optical trap will suffer from aberrations introduced by the microfluidic environment, the effect of the walls of the microfluidic channel on the focusing properties is investigated first. In total, five optical interfaces (see Fig. 2) between different optical materials at 1070 nm [1: Air-BF33 ($n_{\text{rel}} = n_2/n_1 = 1.46$); 2: BF33-Ordyl ($n_{\text{rel}} = 1.08$); 3: Ordyl - H₂O ($n_{\text{rel}} = 0.84$); 4: H₂O - D263 ($n_{\text{rel}} = 1.14$); 5: D263 - H₂O ($n_{\text{rel}} = 0.88$)] have to be traversed by the focused laser beam. Obviously, the first interface leads to the highest refractive index discontinuity and will contribute the most to spherical aberrations. Therefore, ray-optics-based simulations were performed to study the change of the spot radius [geometrical spot size (GEO), root mean square spot size (RMS)] of an ideal focusing system in the presence of additional optical interfaces on the light path. Figure 4 shows three cases for a maximum focusing angle of 55° (NA = 1.1): the ideal case (GEO = 0); a cover slip (thickness 170 μm) introduced 2 mm in front of the focal point (GEO = 15.2 μm); and the layer stack of the microfluidic chip (GEO = 82.1 μm).

The examples shown in Fig. 4 demonstrate the rapid increase of the spot radius when optical interfaces are introduced in front of the focal point of an ideal focusing optics. To estimate how the spot size affects an OT, optical force simulations were performed using a ray-optics-based force simulation routine described in a former publication [9]. A laser power of 50 mW is assumed for the optical trapping of a spherical fused silica particle with 10 μm in diameter. The diagram presented in Fig. 5 shows the optical force acting in the direction of beam propagation along the optical axis (axial or Z direction) for three cases of different spot radii using an annular-shaped intensity distribution, as shown at the top left. The optical forces available for optical trapping are represented by the negative force values (vertical axis in Fig. 5) plotted over the axial dimension. As presented in a previous study [9], optical forces decrease if the spot radius increases, due to spherical aberrations. For the ideal case, a maximum trapping force of 9.4 pN is theoretically achieved. Increasing the spot radius to GEO = 15.2 μm (cover slip) results in a decreased force of 6.9 pN. Additionally, the curve shape becomes less steep

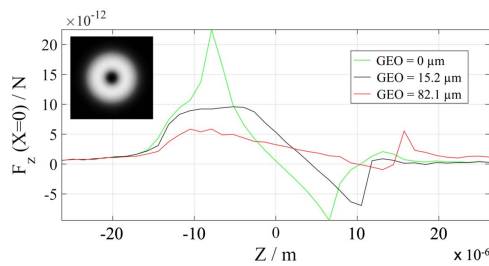


Fig. 5. Axial optical forces (F_z) for an ideal (green) optical trap as compared to optical traps affected slightly (black) and strongly (red) by optical aberrations.

but optical trapping is still possible. When introducing the microfluidic chip, the glass layers cause a spot radius increase of $\text{GEO} = 82.1 \mu\text{m}$, which does not result in an efficient optical trap. The optical trapping force shrinks to 0.9 pN. Since the optical force is directly proportional to the laser power, it is theoretically possible to increase the trapping force by increasing the laser power, but unfortunately, the profile of the force curve (red line) became unfavorable for optical tweezing. The force well is flat, and an additional force maximum arises behind the main optical trap. If the particle is slowed down but slowly leaves the nearby environment of the optical trap, it will be again accelerated away from the trap (second red peak at $Z = 15 \mu\text{m}$), which makes optical trapping by adjusting the laser power practically impossible. This result raises the importance of compensating the introduced aberration, for which two approaches were studied.

Usually, spatial light modulators are applied to compensate for optical aberrations [37–39]. To achieve a cost-efficient optical system with comparatively low alignment effort, we keep the number of optical components small and confront the origin of aberrations already in the optical design process. The first approach was to modify the profile of the outer chip surface, so that the perpendicular incidence of the whole focused ray fan would be ensured. This can be realized using a droplet lens, as can be found for telecommunication applications [40–42] to be assembled on the facet of an optical fiber. For this design, the radius of the curvature and the height have to be specially adapted to the incident ray bundle and the wall layers of the microfluidic channel. Using ray optics we simulated a lens, as visualized in Fig. 6, which significantly reduced the amount of spherical aberrations to a geometrical spot radius of $\text{GEO} = 3.7 \mu\text{m}$. Unfortunately, since the manufacturing tolerances (height or SAG, radius of curvature) of this lens are very narrow, and only a small lateral misalignment of the lens on top of the chip surface will lead to additional aberrations, we preferred a second approach. Nevertheless, due to its reasonable reduction of spherical aberrations, we decided to mention this approach here.

Instead of adapting the surface profile of the chip, the focusing optics of the OT was directly adapted to the first surface of the chip's layer stack by index matching of the OT. The chip was build on top of a Borofloat glass wafer [34] with a refractive index of $n_{1070 \text{ nm}} = 1.462$. Due to its comparatively good manufacturability, PMMA was used for the material of the OTs

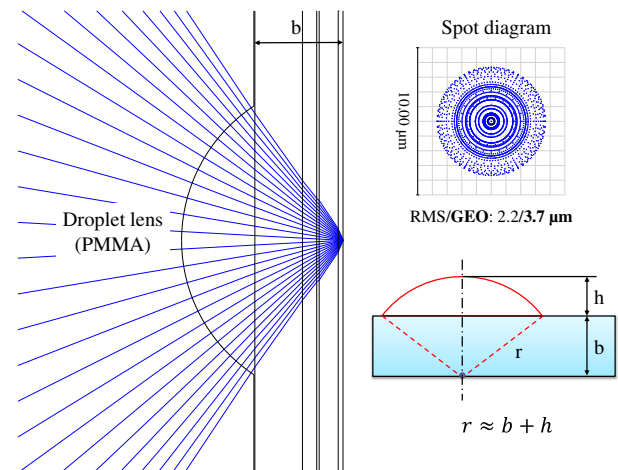


Fig. 6. First and discarded approach for decreasing spherical aberration by use of a droplet lens.

focusing optics. Its refractive index is $n_{1070 \text{ nm}} = 1.482$, which is very close to BF 33 ($n_{1070 \text{ nm}} = 1.462$). As demonstrated in a previous publication [10], a non-spherical but parabolically shaped surface ideally focuses rays, which are incident parallel to its rotational axis, in one focal point located inside the component. We use this geometry as a starting point for the optical design process for the implementation of high-NA focusing to generate the optical trap.

For the beam shaping task we applied a monolithic double axicon to transform a Gaussian TEM_{00} input intensity distribution to a collimated ring distribution, as shown in the top left in Fig. 5. To implement the 90° beam deflection, a right-angle prism made of glass was used. Finally, the high-NA focusing was achieved with a parabolic lens segment made of PMMA, which was directly mounted on top of the chip on the opposite face of the right-angle prism.

B. Design

Based on the concept discussed in the previous section, the OT's initial design (direct contact of high-NA focusing optics with the chip, followed by 90° beam deflection, and Gaussian to ring beam shaping) was adapted to the space restrictions of the magnet bore, and to the microfluidic channel structure as well. As already pointed out in Table 1, a semi-cylindrical volume was available for the optical system. Figure 7 shows the start and the final design of the OT.

A collimated Gaussian beam is incident on the tip of a conical surface representing the first axicon (beam waist located at the axicon tip) and shaped to a ring profile which diverges as it propagates through the optical component. The outer diameter of the first axicon surface is 5 mm with a tip angle of 45° . Due to total internal reflection (TIR) at the surface of the second axicon (cone angle of 8.243°), the ring distribution is collimated. This type of double axicon offers the advantage of a planar-shaped output surface, which can be easily assembled to a second planar component. With a total length of 40 mm for the double axicon, enough space is left for the beam deflecting element. The annular beam leaves the double axicon and immediately enters a right-angle prism. Again, due to TIR

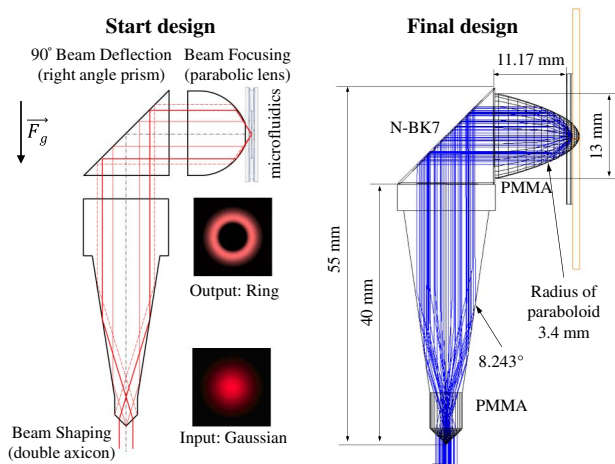


Fig. 7. Initial (left) and final design (right) of the OT.

on the hypotenuse face, the beam is deflected by 90° and leaves the component. In contrast to the double axicon, the right-angle prism will be a standard component (PS915—N-BK7 Right-Angle Prism, Uncoated, 15 mm [43]) made of borosilicate glass [34] and is implemented in the optical design process as a constant set of parameters. Its leg length is 15 mm and thereby matches the outer diameter of the double axicon planar surface.

Once the annular beam leaves the prism, it enters a customized lens, directly mounted on the leg surface of the prism. The lens has a parabolic shape with a planar output surface. That way only a segment of a paraboloid is used. The focal point of this lens is positioned inside the microfluidic channel of the chip. It has been pointed out during the optics design process that the aberrations introduced by the microfluidic channel layer stack only weakly affect the focusing properties if the first interface is index matched. Thus, the focusing lens has a purely parabolic surface (without any correction terms) with a curvature radius of 3.4 mm. If the planar output surface of the parabolic lens is not in contact with another optical material of high refractive index ($n > 1.3$) the light cannot leave the component due to TIR. Its total length is 11.7 mm with an entry diameter of 13 mm and an exit diameter of 5.4 mm. Finally, the fully assembled optical system has a total length of 55 mm, a total width of 27 mm, and its thickness is 15 mm overall.

This design represents the result of the iterative optical design process discussed in Section 2. During the design of the optical elements and the system, a minimum spot radius is achieved by tuning the diameter of the ring distribution and the properties of the parabolic surface of the focusing lens. The optimal inner and outer ring diameters are 4 mm and 9.5 mm, respectively. Rays within this ring will be focused within the smallest achievable spot radius for the present microfluidic chip. This leads to a theoretical spot radius of $\text{GEO} = 5.5 \mu\text{m}$, which is comparable to the droplet lens approach ($\text{GEO} = 3.7 \mu\text{m}$; see Fig. 6), and a significant improvement over the non-adapted system ($\text{GEO} = 82.1 \mu\text{m}$). Simultaneously, the parameters of the optical components are optimized, focusing on a maximum trapping performance

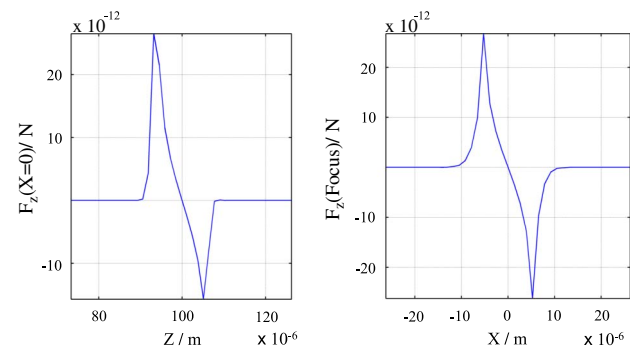


Fig. 8. Simulation of optical forces acting in the axial (F_z , left) and radial (F_x , right) direction on a spherical fused silica particle (10 μm in diameter) suspended in water, using a laser power of 50 mW.

using the optical force simulation. Figure 8 shows the results of the optical force simulation of the final system design presented in Fig. 7 (right).

The maximum optical trapping force theoretically achievable with this system, in the direction of propagation, acting on a spherical fused silica particle (diameter 10 μm) surrounded by water, with a laser power of 50 mW, is 16 pN. For the force acting against gravity, the radial optical forces (acting in the X or Y direction) must be utilized, which are comparatively larger ($F_x = 25$ pN). For a spherical particle made of fused silica the gravitational force is 11.3 pN whereas the buoyancy force will act in the opposite direction with a value of 5.1 pN for a liquid water-based surrounding. Thus, a net gravitational force of 6.2 pN has to be overcome with the optical tweezers, neglecting drag or other force contributions in the microfluidic channel. These results theoretically prove the suitability of the designed trap. After the tolerance analysis, especially the investigation of the effect of manufacturing tolerances (deviation of paraboloids radius of curvature, wrong axicon cone angles, etc.) and alignment tolerances (position mismatch of components to each other) on the optical trap, the design data of the custom components were translated to computer-aided design format for manufacturing.

4. OPTICAL SYSTEM

The optical components (double axicon and parabolic lens) manufactured externally [44] were made of PMMA using an ultra-precision milling process (shape accuracy < 150 nm, surface roughness $R_a < 30$ nm). Figures 9(a) and 9(b) show a photo of the unmounted components. The assembled optical system is shown in Fig. 9(c). Commercially available UV adhesive (NOA 72 [45]) was used to optically and mechanically link the components. Next, we will discuss the interferometrically measured surface profiles of the optical components, followed by the setup for optical trapping.

A. Optical Characterization

In order to assess the efficiency of the assembled optical system we verify the optical quality of the manufactured surfaces. For this we performed measurements using white-light interferometry. It turns out that the tip of the axicon is rugged, which will cause aberrations and intensity losses; see Fig. 10(b). The planar

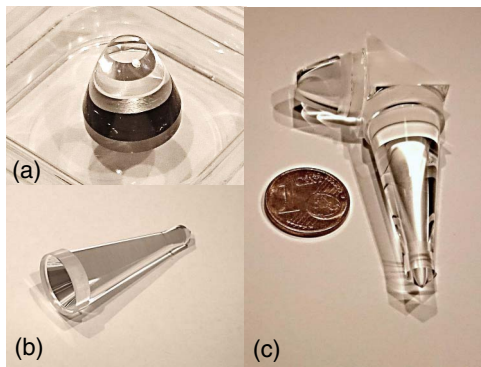


Fig. 9. Photographs of (a) the externally manufactured parabolic lens and (b) the double axicon, both made of PMMA without any additional surface coatings. (c) Optical trapping system comprising a right-angle prism, the double axicon, and the parabolic lens.

output surface of the double axicon is visualized in Fig. 10(a). Since this surface will be directly linked to a leg surface of the right-angle prism, the UV adhesive will partially compensate the tool marks (roughness $R_a \leq 10$ nm) due to its refractive index close to PMMA and glass ($n_{\text{N-BK7}} = 1.51$, $n_{\text{NOA72}} = 1.56$, and $n_{\text{PMMA}} = 1.48$).

Subimages (c) and (d) in Fig. 10 show false color images of the white-light interferometry measurement of the double axicon cone with magnifications of 20 $\times$ and 50 $\times$, respectively.

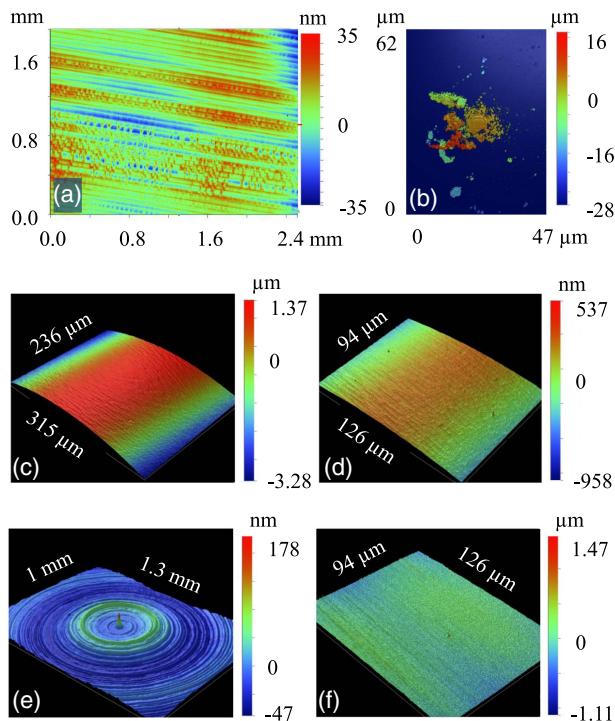


Fig. 10. Profiles recorded by white-light interferometry of the planar surface, the tip, two positions on the second axicon cone of the double axicon (a) ($R_a \leq 10$ nm); (b), (c) ($R_a \leq 30$ nm); (d) ($R_a \leq 30$ nm), and of the planar and parabolic surface of the parabolic lens (e), (f) ($R_a \leq 10$ nm), respectively.

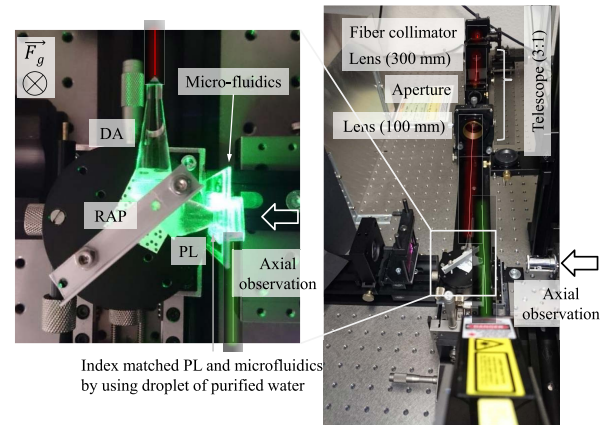
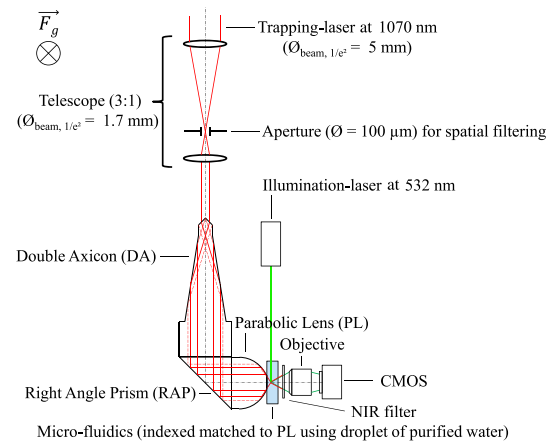


Fig. 11. Schematic (top) and photographic image (bottom) of the optical setup with an inset showing the OT mounted on top of a five-axis platform (top view). (DA, double axicon; RAP, right-angle prism; PA, parabolic lens).

After subtracting the known curvature of the cone from the measurement data, a plane surface with a periodic roughness of $R_a \leq 30$ nm due to tool marks along the radius of curvature appears. The optical function of this conical surface is to collimate the divergent propagation of the ring distribution inside the component by TIR. Such a rough surface will cause an out-coupling of intensity due to a violation of the TIR condition and will finally lead to an additional intensity and thus efficiency loss. However, during the alignment of the component in the trapping laser beam the out-coupling of high-intensity portions of the beam could not be observed nor measured.

The planar input surface of the parabolic lens has an equivalent roughness and comparable tool mark features similar to the planar surface of the double axicon, shown in Fig. 10(a). Also, the surface of the parabolic lens, see Fig. 10(f), shows a similar profile compared to the measurement results of the double axicon cone. Finally, Fig. 10(e) shows the center area of the outer surface of the parabolic lens. A narrow peak of 200 nm in height is left from the manufacturing process. Since this part will be directly connected to the microfluidic chip and index matched, such surface inhomogeneities will barely affect the optical function of the system.

Optical characterization of the surface components shows values for the surface roughness below $R_a = 30$ nm and often below $R_a = 10$ nm, as was advertised by the manufacturer [44]. Therefore, the components were assembled to one monolithic optical system and implemented inside a setup for the investigation of the optical trapping capabilities.

B. Optical Trapping Setup (Demonstrator)

For optical trapping, an optical setup shown in Fig. 11 is used. This setup was utilized in a previous project and is described in detail in [10].

The trapping laser beam (NIR, 1070 nm, continuous wave, $M^2 < 1.2$) leaves the laser source through an optical fiber. By use of a fiber collimator, the beam is collimated and its diameter ($\frac{1}{e^2} = 5$ mm) is adapted to the input aperture of the double axicon with a 3–1 telescope providing spatial filtering by positioning an aperture (100 μm diameter) in its Fourier plane.

The whole OT is mounted on top of a five-axis platform to ensure precise alignment with respect to the incident trapping laser beam as well as to the microfluidics. As a result of the previously performed tolerance analysis, special attention must be paid to the correct alignment of the axicon tip's position and orientation with respect to the incoming laser beam. A lateral offset between both will cause a non-symmetric annular beam profile that can lead to a non-axial symmetric behavior of the optical trap; see, e.g., the results published in [9]. The tilt tolerance of the axicon, however, is even more critical. Small deviations ($\leq 0.1^\circ$) will cause a broadening of the focal spot and lead to a rapid decrease of the trapping performance. Observation of optical trapping in the axial direction (trapping direction) is realized using an objective, NIR filter, and a CMOS camera. To remove the disruption caused by the trapping laser beam during observation, an additional low-power laser (532 nm) is used to illuminate the particles.

5. OPTICAL TRAPPING

In order to experimentally verify the functionality of the developed optical trap, we first calculated the forces acting on the particle to be trapped. This allows us to compare the experimental results with the theoretical predictions performed in Section 4. Since the three-dimensional region of an optical trap influence is limited [volume approximately $(15 \mu\text{m})^3$; see the distance between the maximum and minimum optical forces in Fig. 8], we studied the geometrical conditions of the trapping chamber to estimate the chance to optically manipulate small objects.

A. Preliminary Considerations

As performed in the simulation, we use spherical fused silica particles of 10 μm [46] in diameter to investigate the optical trapping. These particles have a high density and excellent optical properties; trapping such heavy particles represents an important benchmark for possible subsequent experiments with lower-weight polymers or cells.

Considering the buoyancy force acting on such an object, which is placed in a liquid and stationary medium (purified water), a force of 6.2 pN has to be generated by the optical tweezers in order to optically hold the particle. Taking into account a laminar flowing medium that produces drag on

the particle, and an additional movement of the particle due to gravity, the particle possesses kinetic energy, and even higher trapping forces have to be produced to stably trap the particle before it can be held stationary.

For the trapping experiment we built a sample chamber with an optically equivalent wall layer stack (see Fig. 4 right) to the microfluidic channel used for the final system (see Fig. 2 right). The extra sample container enabled us to investigate the influence of the channel wall on the optical trapping and provided a convenient realization of the particle supply. Due to the large volume (height $\times$ width $\times$ thickness = 30 mm $\times$ 9.5 mm $\times$ 4.5 mm) of the sample chamber, the probability to trap a particle is low compared to the microfluidic channel. The particle had to reach the immediate surrounding of the optical trap driven by gravity and drag in order to experience reasonable optical forces that act on the particle and attract it to the focal point; see Fig. 8. The point of zero crossing marks the position of the optical trap (neglecting external forces, such as gravity or drag) and a particle will be accelerated to this point if it is brought close by. In the presence of additional forces, this point will shift accordingly. Simulations show that these trapping forces act on a particle placed within a virtual spherical shell of 20 μm diameter ($\approx 4 \times 10^{-6} \text{ mm}^3$) with the optical trap at its origin. This illustrates the low probability to trap a particle, which moves freely within a cubic volume, 5 mm³ for instance, in terms of geometrical limitation. The simplest approach to increase the probability of trapping is to increase the particle density, and for the moment of trapping, to increase the laser power. Both approaches are pursued in the experiment.

B. Experiment

For the optical trapping experiment the sample chamber was filled with purified water. Next, its wall was moved close to the planar output surface of the optical tweezers by use of a manual precision linear stage. Once the proximity contact was established, both surfaces were index matched using purified water. Therefore, a small droplet of water was released right above the parabolic lens using a syringe. Due to capillary forces, the droplet homogeneously filled the narrow gap between the sample chamber wall and the parabolic lens.

Initially, the trapping laser source was set to a laser output power of 1 W. The particles were introduced to the filled sample chamber by using a syringe filled with a suspension of particles in water. Immediately, the particles started to move along a convection-current-shaped trajectory due to the local heating of the liquid with the focused trapping laser beam. This movement meant that particles moved upward at the center of the chamber and downward at its edges, which finally acted like a continuous particle supply. The optical trap was located in the region of upward flow of particles.

Once a particle moved close to the laser focus it was optically trapped. The optical trapping could be observed using the CMOS camera: the optically retained particle stayed stably at the trap's position, while free particles were moving, as shown in Fig. 12. The laser power was then successively reduced to investigate the minimum laser output power required for optical retention under the given conditions. Within 20 frames (frame rate 15.5 frames per second) free particles (red arrow in Fig. 12) move on average a distance of 500 μm , which resulted

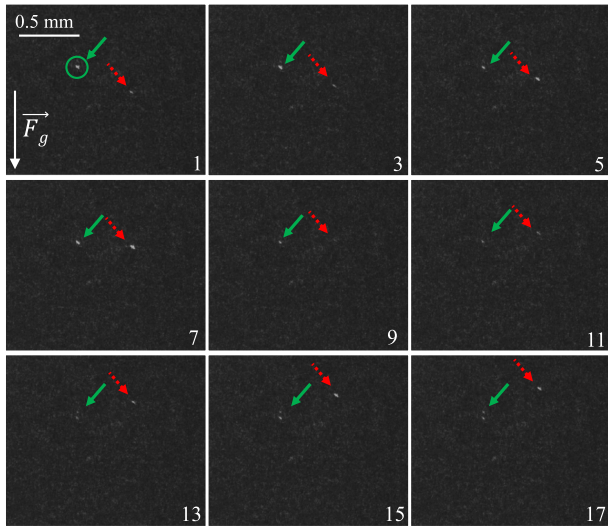


Fig. 12. Optical trapping of fused silica particle (green arrow) and a free particle passing by (red arrow) shown for nine frames recorded with 15.5 frames per second. The green circle marks the position of the optical trap, which was visible as a bright spot during camera positioning tests without the IR filter. Only every second frame is shown to highlight the movement of free particles.

in a particle speed of 0.39 mm s^{-1} . Keeping in mind that the particles moved upward against gravity, we conclude that this movement is due to drag caused by an upward flow of the surrounding medium, at a rate which was faster than the particle speed.

For the present case, the following forces acted on a free particle: gravity [Eq. (1)], buoyancy force [Eq. (2)], and drag [Eq. (3)] due to laminar flow, which was characterized by the Reynolds number ($\text{Re} = 0.087 \ll 1$):

$$F_g = -\frac{4}{3}\pi r^3 g \rho_{\text{particle}}, \quad (1)$$

$$F_b = \frac{4}{3}\pi r^3 g \rho_{\text{H}_2\text{O}}, \quad (2)$$

$$F_{d,\text{laminar}} = 6\pi\eta r(v_{\text{flow}} - v_{\text{particle}}). \quad (3)$$

In all equations, r represents the particle radius ($5 \text{ }\mu\text{m}$), $g = 9.81 \text{ m/s}^2$ is the gravitational acceleration, and ρ is the density ($\rho_{\text{H}_2\text{O}} = 1000 \text{ kg/m}^3$; $\rho_{\text{particle}} = 2201 \text{ kg/m}^3$). The velocities v are given in m/s^{-1} , and $\eta = 0.891 \text{ mPas}$ is the dynamic viscosity of purified water at a temperature of 25°C .

Based on the balance of power, the average velocity of the fluid in the area shown in Fig. 12 can be calculated and results in 0.46 mm s^{-1} . This results in a (non-optical) net force (gravity plus drag) of 32.5 pN , which has to be overcome by the OT in order to stably hold a particle.

Figure 8 (right) shows the trapping forces acting in the axial direction, which are applied in the current setup to trap particles. They show a maximum force of 26.5 pN for a laser output power of 50 mW using an optical tweezers according to the design shown in Fig. 7. Due to the linear relationship of the

laser power and the optical trapping force, a minimum laser power of 61.3 mW has to be applied for optical holding.

In the experiments we investigated a minimum laser power of 170 mW necessary to stably hold a particle over a period of several hours. For lower values of the laser power the particles always left the trap. It has to be taken into account that the laser power was measured at the fiber collimator right behind the trapping laser source. Former experiments [9] have shown that the telescope and its spatial beam filtering as well as the OT itself will cause significant intensity losses. In [9] we determined a minimum intensity loss of 33% due to Fresnel losses and non-ideal (realistic) optical system properties. Since the current system is based on TIR and the trap is formed inside a liquid environment, we used standard ray tracing software to study the power losses. They resulted in a minimum intensity loss of 30% for the ideal system (see Fig. 7) without surface roughness and misaligned components. Therefore, an intensity loss of 40%–50% is a realistic assumption, which leads to a laser power of $85\text{--}102 \text{ mW}$ at the position of the optical trap. This value is still 38%–66% smaller compared to the applied laser power. It shows that probably the applied theoretical model (ray tracer) used to describe the optical system and to calculate optical forces is not accurate enough. In addition, optical aberrations due to manufacturing tolerances (see Fig. 10) and not perfectly aligned optical components, will most probably add additional efficiency losses to the OT which we did not consider within the simulation. Despite these considerations, we demonstrated that the system is suited to stably trap comparably heavy glass particles.

6. CONCLUSION AND OUTLOOK

In this paper we presented the implementation of an optical tweezers for the 3D trapping of small objects inside a sample container/microfluidic channel with a multilayered wall. The layer stack generates several optical interfaces, whose aberrations are addressed and minimized in the optics design. The whole optical system is planned to be integrated in the wide bore of a superconducting magnet. Based on optical force simulations (a geometrical optics approach), we used a novel iterative optical design process to design, optimize, and tolerance the optical tweezers system. Once each individual component was manufactured, we characterized their surfaces using vertical scanning interferometry. Subsequently, the optical system was assembled and integrated in a laboratory setup to experimentally verify the optical trapping ability of the developed OT. Finally, we verified stable optical trapping of fused silica particles with $10 \text{ }\mu\text{m}$ diameter with a laser power of 170 mW .

We targeted an application that requires the mechanical and optical connection of the optical tweezers with a micro-NMR detector that contains a microfluidic channel. The trap was successfully operated on $10 \text{ }\mu\text{m}$ spherical silica particles.

Current work is focusing on the mechanical and optical connection of the optical tweezers with a micro-NMR detector (LiquidVoxel, Voxalytic GmbH) that contains a microfluidic channel. We realize this by using a specially designed mechanical adapter that provides the alignment of the optical system relative to the microfluidic chip to maintain a purified water supply for index matching of the channel and the optical system. For the

investigation of suitable measurement parameters (pulse length, sequence and power, bandwidth, flip angle, gradient, etc.) for the upcoming studies of cells, we will initially use spherical PMMA particles with a diameter of 10 μm . They should provide a clearly distinguishable proton signal when compared to the deuterated liquid environment inside the channel.

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